

THE STRUCTURE OF MATHEMATICAL REALITY

Phase II — Chapter 4

Formal Systems and the Architecture of Proof

Quiz

Self-Study Curriculum

Before you begin

This quiz tests application, not memorisation. Answer in your own words.

Do not quote the lecture back at yourself. If you find yourself writing a phrase you remember reading, stop and rephrase it from your own understanding. A short, precise answer in your own language is worth more than a long transcription.

1. A colleague says: “Godel proved that math is just subjective, you can choose any axioms you want and get different results.” What is right about this, and what is importantly wrong?

2. Can a formal system prove all the things that are true within it? Under what conditions?

3. What is the structural difference between “this problem is unsolved” and “this problem is independent of our axioms”?

4. You are building a predictive model and you find that two models with different assumptions make the same predictions on all available data. Is this analogous to any concept from this chapter? Which one?

After the quiz

For each answer, ask yourself: “*Did I explain the **why**, or did I just state the **what**?*” If you stated a fact without explanation, go back and add the reasoning.

Date: _____ **Questions where I want to go deeper:**
